Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Page screening in sparse attention can discard useful distant tokens before refinement, making its scoring representation a critical part of candidate selection. We study a near–far cascade whose page screening and token refinement share 32 low-frequency RoPE coordinates, arranged as 16 complete rotation pairs. Separate regional page quotas retain candidate access to near and far history, while final attention uses the selected original-dimensional keys and values. On five LongBench tasks with Qwen3-8B, page/token scoring at 32/32 dimensions (sub/sub) averages 47.588; nominal 128/32 (full/sub) and nominal 32/128 (sub/full) configurations score 47.910 and 47.666. Incomplete configuration binding limits these differences to descriptive comparisons. A separate synthetic study reports no selector-latency reduction for sub/sub relative to full/sub. The results motivate evaluating compact scoring jointly with selection quality and preparation-inclusive cost, without establishing a joint quality–latency advantage.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context sparse attention must retain useful history while keeping token selection cheaper than the attention work it removes. A two-level selector makes this objective depend on two decisions. Page screening determines which tokens can reach refinement: a rejected page cannot be recovered at the next stage. The scoring representation determines the arithmetic per comparison, but representation preparation, ranking and memory access also contribute to latency. We therefore study scoring dimensionality and candidate allocation as distinct design choices.

We investigate a shared low-frequency representation for a near–far selection cascade. Each layer and KV head uses one set of complete RoPE coordinate pairs for page screening and token refinement. The cascade assigns separate page quotas to recent unprotected history and the remaining history, allowing distant candidates to enter refinement even when the near region dominates page selection. Original-dimensional Q/K/V determine attention over the resulting token set. Sharing the scoring view reduces representational duplication in the design; regional quotas control candidate access. Neither mechanism alone guarantees lower latency or unchanged task quality.

The design raises three questions: whether compact coordinates retain useful candidates, whether regional quotas improve retrieval at a fixed final token budget, and whether attention savings exceed representation preparation and selection costs. Figure 1 separates retrieval demand from scoring fidelity: distance-conditioned attention describes where full-reference mass lies, while equal-width paired-coordinate comparisons measure the mass retained by selected tokens. These diagnostics remain pending; the available task scores and local timings neither isolate stage-specific effects nor establish a joint quality–latency advantage.

II. Related Work

Query-aware retrieval methods such as Quest, SparQ and HISA reduce the active attention set through approximate selection [1], [2], [3]. AsyncTLS combines group-shared page/token selection, an average of per-query-head probabilities in refinement, and full-dimensional final attention [4]. Our design builds on this cascade structure and studies a different representation and allocation choice: a common set of low-frequency RoPE coordinate pairs across both stages, coupled with separate near and far page quotas. AsyncTLS instead uses full-dimensional page scoring, calibrated refinement channels and asynchronous execution. These distinctions separate the representation question from the benefits of a particular execution schedule.

KV compression and placement methods address complementary memory-capacity and data-movement constraints [5], [6], [7]. The method studied here preserves original keys and values and changes their selection. Consequently, it does not inherit a reduction in original-KV cache capacity from the compact scoring representation.

III. Near–Far Selection in a Shared Subspace

The selector first constructs a shared low-frequency scoring view, then partitions unprotected history into near and far regions. Each region undergoes page screening followed by token refinement; the resulting indices gather original-dimensional keys and values for attention (Fig. 2). This separation allows the scoring representation to change while retaining original-dimensional operands for attention over the selected tokens.

A. Shared low-frequency coordinates

For Qwen3-8B, the evaluated representation selects post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. These are 16 complete low-frequency rotation pairs, rather than a contiguous suffix or a learned projection. Their slower positional phase variation motivates examining them as a compact scoring view; it does not imply greater retrieval fidelity, which depends on the query and key content. Scoring Q/K tensors are physically compact at width 32. With coordinate embedding E_J, the scoring views are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. A layer and KV head prepare one key view for both regions and both stages (Fig. 2; Supplementary Fig. S1). Query heads retain their own scores. Original keys and values remain available at full width for final attention and later queries; this design does not reduce original-KV cache capacity.

For a fixed token-independent coordinate map and row selector R, $R(KE_J) = (RK)E_J$. The same compact key view can therefore serve both regions without changing the coordinate-selection operation. Page summaries and quantize–dequantize token keys remain

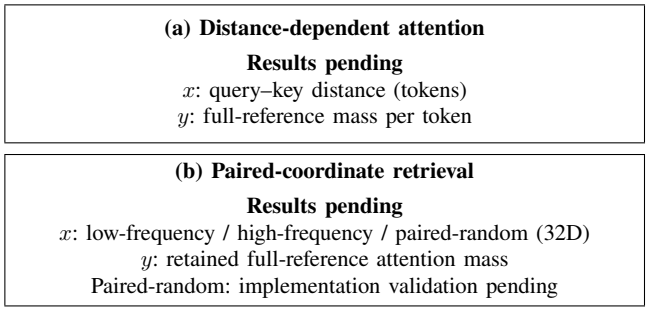


Fig. 1. Diagnostic views of retrieval demand and coordinate selection (measurements pending). Distance-bin mass is normalized by the number of valid tokens, with protected tokens reported separately. Frequency comparisons use the same Q/K, candidate support and budgets.

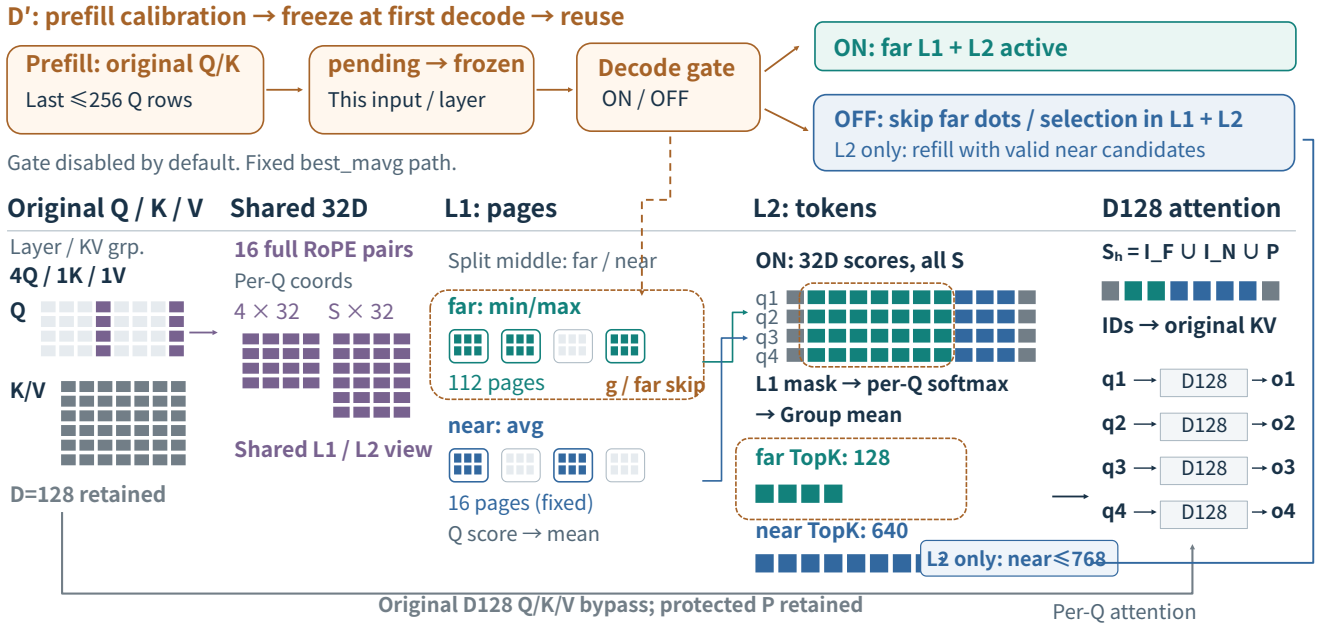


Fig. 2. Shared-subspace near–far selection. Both stages use the same RoPE coordinate pairs; selected IDs gather original-dimensional K/V, while each query uses its original-dimensional Q for final attention. The optional prefill-informed gate suppresses far selection and reallocates unused token quota within the retained near pages. The ordinary refinement implementation computes full-history scores before applying the page mask.

distinct derived representations. Token appends, changing region boundaries and updates to the coordinate map require corresponding maintenance of these representations and their logical IDs. This maintenance, together with original-KV gathers, determines whether representational reuse translates into a shorter execution path.

B. Regional page and token budgets

At decode position t, protect the prefix and recent window,

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of the unprotected middle history; far is its complement. Near is therefore distinct from the protected sliding window. The near boundary follows fraction α of unprotected history under minimum-page and alignment rules. The unprotected middle has length $\max(0, S-|\mathcal{P}_t|)$; defaults are specified in the experimental setting.

Near uses average-key page scores. Far uses sign-aware extrema in the shared space. For each nonempty region-intersected page b, define $m_{b,j} = \min_{i \in b} z_{i,j}$ and $M_{b,j} = \max_{i \in b} z_{i,j}$. Quest's extrema construction gives

$$U_b(z_q) = \sum_j (z^+_{q,j} M_{b,j} + z^-_{q,j} m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. This bounds unquantized shared-space dot products, not original-dimensional logits or separately quantized scores. Extrema can come from different tokens; a top-page shortlist can still miss important tokens. Average-key scores do not inherit this bound.

With $K_1 = 128$ and $\beta = .125$, near and far receive 16 and 112 L1 pages. Regional reservation allocates access, not a quality guarantee: a far page can displace a useful near page. A later token quota cannot recover a token after L1 discarded its page. This distinguishes regional L1 screening from splitting one global shortlist after selection.

For each KV head h, let $\mathcal{A}_{R,h}$ contain valid tokens from the retained pages of region R. For the supported budget domain $K \ge |\mathcal{P}_t|$, the middle cap is $K_{\text{mid}} = K - |\mathcal{P}_t|$. In the source-derived sufficient-length example, a per-head total cap $K = 1024$ leaves 768 middle tokens (Supplementary Fig. S2). In the normal sufficient-capacity path, $\gamma = .625$ acts on near-page capacity, yielding $16 \cdot 64 \cdot .625 = 640$ near tokens and 128 far tokens. Short histories or insufficient candidates can reduce these realized counts. Protection joins selected IDs once; padding is excluded and duplicates consume no extra quota.

C. GQA selection and original-dimensional attention

For 32 query heads and 8 KV heads, each group contains four queries. L1 averages their page scores before regional TopK. Ordinary L2 scores the full history for each query, applies the shared L1 mask, and averages the independently normalized query probabilities within each KV group. This probability aggregation differs from applying softmax to averaged logits. Sharing coordinates therefore preserves query-specific distributions, while full-history scoring means that a smaller page shortlist does not itself reduce L2 dot-product work.

The group's unique selected set $\mathcal{S}_{t,h}$ combines protection and regional winners under the total cap. All queries g in that group use these IDs, but recompute their own full-dimensional attention:

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Merging IDs before this softmax avoids adding separately normalized regional outputs [8]. Retained logits use original KV, but restricting the denominator can change attention output. Shared IDs provide a common gather set; realized KV traffic depends on the kernel's load and reuse behavior.

TABLE I
Sub/sub settings: M denotes saved manifest, C recorded-source inference. K counts unique final tokens including protection.

Parameter	Setting / evidence
Final token cap K	1024 (M)
Page size; L1 page cap K_1	64; 128 (M)
α, β, γ	.25, .125, .625 (M)
L1/L2 scoring coordinates	32/32 (C)
Prefix / recent protection	128/128 (C)
Prefill; decode layers	dense (M); all sparse (C)
L2 scoring support	full history before mask (C)

D. Prefill-informed far gating

Far-history selection incurs work at every decode step, even when a layer assigns little attention to that region. D′ tests whether prefill observations can identify layers where far selection can be omitted. For each input and layer, it assigns OFF when mean far-attention mass observed during prefill is below ϵ, reusing this decision during decoding. Otherwise it retains far selection. The feature is disabled by default, leaving ordinary near–far selection active. On the supported path, OFF skips both far scoring stages and their selections, while unused L2 quota refills eligible near tokens without expanding the near-page budget.

We implement this extension for a single sequence; Supplementary Note S5 gives the observation boundary and state transitions. Its threshold controls a prefill statistic rather than subsequent attention error. The gate is therefore an approximation policy whose effect on quality and total cost must be evaluated separately from the default ungated method.

IV. Experimental Setting

The sub/sub configuration uses Qwen3-8B and the settings in Table I. Its saved manifests specify dense prefill and shared KV-head selection. The recorded producer code supplies the 128-token prefix and recent-window constants and the 32-coordinate view; these are source-derived settings rather than per-step runtime verification. The selector uses 16-level quantize–dequantize values in floating-point tensors, so its execution is distinct from packed INT4 arithmetic. Other result families and unresolved settings are listed in Supplementary Note S7.

Representation and quality. Dense FullKV retains the full attention support and is the quality reference. Full/full instead retains sparse selection with full-dimensional scores in both stages. A controlled L1/L2 factorial comparison requires all four modes with fixed scoring rules, protected sets and model settings; the descriptive scores below do not complete that comparison. For sparse baselines, fairness aligns the final deduplicated token cap K, including protection, on the same model, inputs, decoding phase and dense-layer policy. Internal candidate-page quotas remain method parameters and need not match across methods; their scoring and memory costs are included. A matched comparison must report realized token counts, page rounding and per-query-head versus shared-KV-head selection separately (Supplementary Note S7).

Comparison design. Representation reuse and regional allocation have different controls. Fixed-ID replay holds the selected tokens and attention computation fixed while changing representation construction. An equal-budget regional/global comparison instead changes page allocation under a single page scorer across the entire history. Using near-average and far-extrema scores only in the regional arm would confound allocation with scoring. For the optional gate, far-on, off-refill and off-no-refill distinguish the removal of far computation from changes to the near selection. These controls separate the mechanisms of the method.

(a) Complete request cost

Matched measurements pending

x: actual input length (tokens)
y: request wall time (ms), including preparation

(b) Decode cost

Matched measurements pending

x: actual input length (tokens)
y: time per output token (ms)
Same model, inputs, output length and token budget

Fig. 3. Complete request and decoding cost (measurements pending). Both panels use the same workload as the quality comparison. Request time includes preparation; per-token time averages returned-token intervals after the first output.

Dual-32 near–far selection (sub/sub)

Matched FullKV comparison pending

Reference: dense FullKV on the same inputs
$\Delta Q = Q_{\text{sub/sub}} - Q_{\text{FullKV}}$ (points)
Task-wise and aggregate paired uncertainty: pending

Fig. 4. Quality of sub/sub relative to dense FullKV (matched measurements pending). The intended quantity is the task-score difference on identical inputs, with document-paired uncertainty. Full/full is a representation control within sparse selection, not the dense reference.

Cost and uncertainty. Complete request time includes representation preparation, maintenance, calibration, selection, KV gathers, attention and synchronization, with overlapping work counted once. TTFT measures time to the first returned token; TPOT averages subsequent return intervals. A task-quality/latency comparison requires the same model, inputs, generation settings and scoring procedure. Uncertainty in task-score differences must account for repeated questions from the same source document. Non-inferiority additionally requires an acceptable-loss margin specified before examining confirmation results. The present analysis reports representation scores and local selector timings separately; it does not estimate a joint end-to-end quality–latency frontier. Supplementary Note S6 records the experiment identities and reproducibility limits.

V. Preliminary Results

Reported configuration scores. Across HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], the mean scores are 47.588 (sub/sub), 47.910 (nominal full/sub) and 47.666 (nominal sub/full). Relative to sub/sub, the latter means differ by 0.322 and 0.078 points; sub/full is 0.244 points below full/sub. The nominal full/sub and sub/full runs are not verified as controlled changes to page- or token-scoring dimensionality. Their score differences are therefore descriptive and do not isolate either stage's effect (Supplementary Notes S4 and S7). None is dense FullKV.

The sub/sub–FullKV contrast in Fig. 4 remains unmeasured under verified matching conditions. The three-arm means therefore cannot establish preservation of dense-attention quality. Task-level bootstrap summaries describe variation across these five tasks, rather than a document-paired non-inferiority test.

Local selector cost. A separate synthetic experiment uses single-layer, batch-one, single-query QKV at length 64K. Its two-round steady-state summaries are 1.3502 ms for full/sub, 1.3598 ms for sub/sub and 1.4634 ms for sub/full. The shared-subspace selector thus has no lower reported time than full/sub in this setting. This ordering does not identify a limiting stage, and the unresolved sampling-protocol discrepancy prevents a reliable interpretation of small timing differences (Supplementary Note S6). The operator timings are not paired with the LongBench quality runs and do not measure complete-request cost.

Limits of protection and prefill gating. The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects. Prefill statistics also fail to bound later far-region demand: a record with prefill far-attention mass below .05 reaches .1678 during decoding. Relative to the reused 47.910 reference, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each arm, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity. The complete scoring configuration is not identified in these gate summaries, so the results cannot be attributed to the gated dual-32 method (Supplementary Note S6).

Scope. The results describe configuration scores and the limitations of prefill-based gating on the studied Qwen3 configurations. They do not establish non-inferiority to dense FullKV, the isolated benefit of regional page quotas, or lower complete-request cost. The available 128K RULER records [10] lack a comparable identified sparse arm and are excluded from quantitative comparison. Conclusions are consequently limited to the reported configurations and workloads.

VI. Conclusion

We studied a shared low-frequency RoPE representation for near–far page screening and token refinement. The five-task sub/sub mean is 47.588, compared with 47.910 and 47.666 for the nominal full/sub and sub/full configurations. Incomplete configuration binding precludes attributing these gaps to an individual selector stage. The separate synthetic timing study reports no latency advantage for sub/sub. A competitive quality–cost trade-off must therefore be established using matched final token budgets and preparation-inclusive request measurements, with acceptable quality degradation specified before evaluation.

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OpenAI language models accessed through Codex assisted manuscript drafting and revision, Chinese translation, figure and LaTeX code generation throughout, source checking and arithmetic. The interface did not expose model-version identifiers.

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